THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — SCIENCE · HARD LEVEL CHAPTER TEST SET 2

Chapter 9: **The Amazing World of Solutes, Solvents and Solutions** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. How much water must be added to 100 g of a 40 per cent solution to dilute it to 10 per cent? 1

(a) 300 g **(b)** 30 g **(c)** 400 g **(d)** 100 g

2. A blue solution turns colourless on adding zinc. The solute is probably: 1

(a) ethanol **(b)** sugar **(c)** copper sulphate **(d)** sodium chloride

3. Assertion (A): Fish in a pond die in summer when the water gets very warm. Reason (R): The solubility of oxygen in water decreases at high temperature. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

4. A 100 g sample of saturated solution at 80 degrees Celsius contains 40 g of solute. It is cooled to 20 degrees Celsius where solubility is 25 g per 100 g of water. Mass of water in the sample and of crystals formed: 1

(a) 40 g water and 25 g crystals **(b)** 60 g water and 15 g crystals
(c) 100 g and 40 g **(d)** 60 g water and 25 g crystals

5. A student weighs a beaker of water and sugar before and after dissolving. The mass: 1

(a) remains the same **(b)** decreases **(c)** is zero **(d)** increases

6. Assertion (A): A saturated solution, on cooling, deposits crystals. Reason (R): The solubility of most solids decreases as temperature decreases. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

7. A mixture of fine sand and salt in water is filtered; the filtrate is evaporated. What remains in the dish? 1

(a) water **(b)** sand and salt **(c)** sand **(d)** salt

8. Honey is a solution of: 1

(a) sugars in water **(b)** oil in water **(c)** gases **(d)** salt in oil

9. Assertion (A): Milk shows the Tyndall effect. Reason (R): Milk is a true solution. 1

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

10. In which situation would the rate of dissolving of salt be the least? 1

(a) powder in hot water with stirring
(b) powder in cold water with stirring
(c) coarse crystals in cold water without stirring
(d) coarse crystals in hot water

SECTION — B (Very Short Answer Type) — 4 × 2 = 8

Answer in about 20–30 words each.

11. Why does an unopened soda bottle show no bubbles but opening makes them appear? 2

12. A student claims 'water is a solute in a solution of alcohol in water when water is less'. Comment. 2

13. A solution of 8 per cent contains 24 g of solute. Find the total mass and mass of solvent. 2

14. Explain why a mixture of salt and sugar cannot be separated by simple filtration. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. Explain why the boiling of the soft-drink bottle caps is dangerous. 3

16. Explain the fall in solubility of gases when water is heated using particle ideas. 3

17. How would you separate a mixture of sand, salt and camphor? 3

OR

Describe an experiment to show the presence of dissolved gas in tap water. 3

18. How is dissolving different from mixing sand in water in terms of the particles involved? 3

OR

A student mixes 15 g salt with 85 g water. Another mixes 15 g salt with 185 g water. Compare the concentrations. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Explain the significance of the Tyndall effect and how it distinguishes solutions, colloids and suspensions; give an application in daily life. 5

OR

Compare solution, suspension and colloid with respect to nature, particle size, filtration, settling, Tyndall effect and give two examples of each. 5

20. Discuss how the particle model explains dissolving, dilution, saturation and the effect of temperature on solubility. 5

OR

A student wants to prepare 500 g of 4 per cent salt solution. Describe the procedure, calculating the mass of salt and water, and explain how to check it is not saturated. 5

END OF QUESTION PAPER — Total: 40 Marks